AI Risk-Level Matrix

A Decision-Support Framework for Evaluating Educational Risk in AI-Assisted Academic Work

Version 1.0

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About this Framework

The **AI Risk-Level Matrix** provides a structured framework for evaluating the educational significance of AI-assisted academic work. Rather than classifying AI technologies as acceptable or unacceptable, it encourages educators to consider the educational function performed by AI in relation to the learning outcomes being assessed.

Building upon the principles established in *From Assistance to Authorship*, the matrix distinguishes between supportive AI use that enhances learning and substitutive AI use that may reduce confidence that submitted work represents the student's own intellectual achievement. Educational risk is therefore determined by the relationship between AI assistance, assessment purpose, disciplinary expectations, and professional judgement rather than by the technology itself.

The framework is intended to support lecturers, programme leaders, assessment designers, academic integrity professionals, quality assurance teams, and external examiners in making more consistent, transparent, and educationally defensible decisions. It complements the wider AI Governance Framework Series by providing a common language for discussing educational significance across assessment design, marking, moderation, disclosure, and institutional governance.

Executive Summary

Universities increasingly recognise that not all forms of AI assistance present the same educational concerns. Treating every use of AI as either acceptable or unacceptable oversimplifies a much more complex reality.

The AI Risk-Level Matrix provides a structured approach for evaluating the **educational significance** of AI use rather than the technology itself.

The framework classifies AI activities according to the extent to which they influence the intellectual work that an assessment is intended to evaluate. This enables educators to distinguish between supportive uses of AI that enhance learning and substitutive uses that may undermine authentic assessment.

Importantly, the matrix does not determine whether misconduct has occurred. Nor does it replace institutional policy or professional judgement. Instead, it provides a common language for discussing educational risk across assessment design, marking, moderation, and academic integrity processes.

The framework complements the companion documents by informing assessment design, disclosure expectations, oral verification, and institutional decision-making.

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1. Introduction

One of the greatest challenges in AI governance is inconsistency.

Two lecturers may respond very differently to identical AI use because they evaluate the technology rather than its educational significance. One may regard any AI assistance as problematic, while another may view the same practice as entirely acceptable.

This inconsistency often arises because discussions focus on AI tools rather than the educational functions those tools perform.

The AI Risk-Level Matrix addresses this problem by shifting attention from technology to learning.

The relevant question is not:

Which AI tool was used?

Instead, the more educationally meaningful question is:

What intellectual work did the AI perform?

Risk is therefore determined by the relationship between AI assistance and the learning outcomes being assessed.

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2. What Is Educational Risk?

Within this framework, educational risk refers to the likelihood that AI assistance may reduce confidence that the assessment provides valid evidence of the student's own learning.

Educational risk is **not** synonymous with academic misconduct.

Low-risk AI use may be entirely appropriate.

Higher-risk AI use may also be acceptable in some educational contexts where those intellectual activities are not being assessed.

Consequently, risk should always be interpreted in relation to the assessment purpose.

The matrix therefore supports professional judgement rather than automatic classification.

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3. Principles of Risk Evaluation

The framework is guided by four principles.

Educational Alignment

Risk depends on the learning outcomes the assessment intends to measure.

Functional Evaluation

Risk is determined by the educational function performed by AI rather than the software used.

Proportionality

Institutional responses should be proportionate to educational significance.

Professional Judgement

The matrix informs decisions but does not replace academic expertise.

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4. The AI Risk-Level Matrix

Figure 1. AI Risk-Level Matrix.

A decision-support framework classifying AI-assisted activities according to their educational significance in relation to assessment purpose, learning outcomes, and professional judgement. The matrix distinguishes educational risk from academic misconduct and supports proportionate institutional responses.

The matrix classifies AI-assisted activities according to their potential impact on authentic assessment. The objective is not to prohibit particular AI tools but to evaluate whether AI has performed intellectual work that the assessment intended the student to demonstrate.

Risk should therefore be interpreted in relation to the assessment context, learning outcomes, and disciplinary expectations.

The matrix identifies four broad levels of educational risk.

Level 1 – Low Educational Risk

Supportive AI Functions

Low-risk activities primarily improve the presentation, accessibility, or technical quality of academic work without contributing substantially to disciplinary reasoning.

Typical examples include:

- spell checking;
- grammar correction;
- punctuation;
- formatting;
- reference formatting;
- accessibility support;
- transcription;
- language translation where language proficiency is not the learning outcome.

These activities are comparable to many existing digital writing tools and generally present limited educational concern.

Educational Response

Normally:

- permitted;
- minimal disclosure where required;
- no additional verification.

The central question remains whether the student's own thinking is preserved.

Level 2 – Moderate Educational Risk

Organisational and Developmental Support

At this level, AI begins to influence how students organise and develop their work while leaving substantive academic judgement largely under student control.

Typical examples include:

- brainstorming ideas;
- generating possible research questions;
- producing outlines;
- organising literature;
- suggesting headings;
- improving coherence;
- proposing alternative structures.

These activities may contribute meaningfully to learning when students continue to exercise independent judgement regarding the ideas ultimately adopted.

Educational Response

Normally:

- permitted with appropriate guidance;
- disclosure recommended or required according to institutional policy;
- no routine verification.

Educators should encourage students to evaluate rather than simply accept AI suggestions.

Level 3 – Elevated Educational Risk

AI-Assisted Academic Writing

At this level, AI contributes directly to sections of academic work that demonstrate disciplinary understanding.

Examples include:

- drafting paragraphs;
- rewriting substantial sections;
- summarising literature;
- generating explanations;
- producing case analyses;
- preparing extended responses.

Although students may subsequently edit AI-generated material, these activities increasingly influence the academic work that educators are attempting to evaluate.

Educational judgement therefore becomes more important.

Educational Response

Normally:

- explicit disclosure expected;
- educators review whether learning outcomes remain demonstrated;
- oral verification may be appropriate in selected cases;
- assessment design should be considered carefully.

These activities are not automatically unacceptable.

Their appropriateness depends upon assessment purpose.

Level 4 – High Educational Risk

Intellectual Substitution

High-risk activities occur where AI performs intellectual work that students are expected to demonstrate independently.

Examples include:

- interpreting research findings;
- constructing analytical arguments;
- developing theoretical positions;
- writing discussions;
- generating conclusions;
- producing complete assessment responses representing independent scholarship.

Where these activities constitute the intended learning outcomes, confidence that the submitted work reflects authentic student achievement may be substantially reduced.

Educational Response

Possible responses include:

- comprehensive disclosure;
- educational discussion;
- oral verification;
- assessment review;
- institutional procedures where appropriate and supported by evidence.

Importantly, the matrix does not classify these activities as misconduct.

Rather, it identifies situations requiring the greatest professional judgement.

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5. Risk Is Context Dependent

The same AI activity may represent different levels of educational risk depending upon the assessment.

For example:

Example 1

AI-assisted translation.

Language assessment:

High educational significance.

Engineering report:

Often low educational significance.

Example 2

AI-generated computer code.

Programming assessment:

Potentially high educational risk.

Business planning assignment requiring data visualisation:

May represent acceptable technical assistance.

Example 3

AI-generated summaries.

Research methods assessment evaluating literature synthesis:

Elevated educational risk.

Preparatory reading before class discussion:

Generally low educational risk.

The matrix therefore rejects universal classifications of AI use in favour of context-sensitive educational judgement.

Educational context always matters.

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6. Relationship Between Risk and Institutional Response

Educational risk should inform rather than determine institutional responses. Professional judgement and institutional policy remain essential.

Instead, it informs the proportionality of educational responses.

Risk Level	Typical Institutional Response
Low	Routine assessment

Risk Level	Typical Institutional Response
Moderate	Guidance and disclosure
Elevated	Professional review and possible clarification
High	Educational review and, where justified by evidence, formal procedures

This graduated approach encourages consistency while avoiding unnecessarily punitive responses.

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7. Applying the Matrix

The matrix supports decision-making throughout the assessment process.

Assessment Design

Educators can evaluate whether assessment tasks unintentionally encourage substitutive AI use.

Student Guidance

Disclosure expectations may be aligned with educational risk.

Marking

Markers can interpret AI disclosures more consistently.

Oral Verification

Verification may be informed by the educational significance of disclosed AI assistance rather than assumptions regarding particular software.

Moderation

Programme teams may use the matrix to improve consistency across modules.

Academic Integrity

The matrix provides a common language supporting evidence-based discussion while recognising that final decisions remain subject to institutional policy.

Programme teams should periodically review how the matrix is being interpreted across modules to ensure comparable educational situations are approached through consistent reasoning. Such review supports fairness while recognising legitimate disciplinary variation.

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8. Worked Examples

Case Study 1

Student uses AI to correct grammar before submission.

Risk

Low.

Educational response

Routine assessment.

Case Study 2

Student uses AI to develop an essay outline before writing independently.

Risk

Moderate.

Educational response

Disclosure where required.

Case Study 3

Student asks AI to draft the literature review before rewriting it substantially.

Risk

Elevated.

Educational response

Professional judgement regarding whether learning outcomes remain demonstrated.

Case Study 4

Student submits an AI-generated discussion chapter with minimal modification.

Risk

High.

Educational response

Educational review and, where appropriate, institutional procedures.

The decision should always be based on the totality of available evidence rather than AI use alone.

Professional development should encourage dialogue and calibration rather than uniform decision-making.

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9. Lecturer Decision Checklist

The AI Risk-Level Matrix is intended to support consistent and evidence-informed professional judgement. It should not be used as a scoring instrument or a substitute for institutional policy.

The following checklist may assist educators when evaluating AI-assisted assessment.

Step 1 — Identify the Learning Outcome

☐ What knowledge, skills, or capabilities is this assessment intended to evaluate?

☐ Which aspects of the assessment require independent intellectual judgement?

☐ Would AI performing this task undermine the intended learning outcome?

Step 2 — Identify the AI Function

☐ What did the AI actually do?

☐ Was AI used primarily for technical support, organisation, drafting, or analytical reasoning?

☐ Which intellectual activities remained the student's responsibility?

Step 3 — Determine the Educational Risk

☐ Low

☐ Moderate

☐ Elevated

☐ High

Risk should be determined by educational significance rather than the software used.

Step 4 — Consider Context

☐ Is this use appropriate for the discipline?

☐ Does institutional policy permit this form of AI assistance?

☐ Has the student disclosed AI use appropriately?

☐ Does the assessment still provide credible evidence of learning?

Step 5 — Select the Educational Response

☐ No further action

☐ Provide feedback

☐ Clarify AI use

☐ Conduct oral verification

☐ Review assessment design

☐ Refer through institutional procedures (only where justified by evidence)

Figure 2. Lecturer Decision Checklist. A structured decision-support tool to guide educators through the evaluation of AI-assisted academic work. The checklist encourages evidence-based professional judgement by considering the intended learning outcomes, the educational function performed by AI, the level of educational risk, contextual factors, and the most appropriate educational response. It supports consistent decision-making while recognising that the framework informs rather than replaces institutional policy and academic expertise.

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10. Programme-Level Applications

Although the matrix may support individual assessment decisions, its greatest value lies at programme level.

Programme teams may use the matrix to review assessment portfolios and identify patterns across modules.

Questions include:

- Are expectations regarding AI use consistent across the programme?
- Do different modules classify similar AI activities differently?
- Are disclosure requirements proportionate?
- Do assessments collectively generate sufficient evidence of authentic learning?
- Are moderation decisions consistent across teaching teams?

Programme-level discussion reduces inconsistency and promotes coherent governance.

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11. Staff Development

Successful implementation depends upon staff confidence as much as institutional policy.

Professional development should therefore focus on helping educators:

- distinguish educational risk from academic misconduct;
- interpret disclosure statements consistently;
- design assessments that reduce unnecessary educational risk;
- conduct proportionate oral verification where appropriate;
- exercise professional judgement confidently.

The objective is not to produce identical decisions in every case, but to ensure that similar cases are approached through comparable reasoning.

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12. Relationship with the Companion Framework Suite

The AI Risk-Level Matrix functions as the principal decision-support framework within the AI Governance Framework Series.

Its role differs from the companion documents.

- **From Assistance to Authorship** establishes the conceptual principles for evaluating AI-assisted academic work.
- **AI-Integrated Assessment Integrity Lifecycle** embeds those principles throughout the assessment process.
- **AI Disclosure Framework** supports transparent reporting of AI assistance.
- **Oral Verification Toolkit** provides procedures for confirming intellectual ownership where additional evidence is required.
- **Assessment Authenticity Checklist** supports assessment design that preserves meaningful evidence of learning.

The AI Risk-Level Matrix connects these frameworks by informing professional judgement across each stage of assessment.

Rather than prescribing institutional decisions, it provides a common vocabulary for discussing educational significance.

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13. Limitations

The matrix is intended to guide educational judgement rather than classify all AI use definitively.

Educational risk is influenced by numerous contextual factors, including:

- assessment purpose;
- disciplinary expectations;
- programme learning outcomes;
- institutional regulations;
- professional accreditation requirements.

Consequently, identical AI activities may reasonably be interpreted differently in different educational contexts.

The matrix should therefore be applied alongside institutional policy and professional expertise.

It is a conceptual and practice-informed framework. It has not yet undergone formal empirical validation as a decision-support instrument.

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14. Conclusion

AI governance requires more than identifying which technologies students use.

It requires understanding how those technologies influence learning.

The AI Risk-Level Matrix shifts attention from software to educational function. In doing so, it encourages more consistent, proportionate, and educationally meaningful discussions about AI-assisted assessment.

The matrix does not determine whether AI use is acceptable.

Nor does it replace institutional policy.

Instead, it provides educators with a structured approach for evaluating the educational significance of AI assistance while preserving flexibility for disciplinary judgement and local implementation.

Used alongside the companion frameworks, it supports assessment systems that remain focused on authentic learning rather than technological surveillance.

Framework Status

Framework type: Decision-support framework

Development status: Conceptual and practice-informed

Validation status: This framework is conceptual and practice-informed. It has not yet undergone formal empirical validation as a decision-support instrument.

Primary users

- Lecturers
- Programme leaders
- Assessment designers
- Academic integrity staff
- Quality assurance teams

- External examiners

Companion frameworks

Framework	Contribution
From Assistance to Authorship	Conceptual foundation
AI-Integrated Assessment Integrity Lifecycle	Assessment implementation
AI Disclosure Framework	Transparency
Oral Verification Toolkit	Verification of authorship
Assessment Authenticity Checklist	Assessment design
Institutional AI Governance Guide	Institution-wide implementation

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15. Version History

Version	Date	Description
1.0	July 2026	Initial public release
